

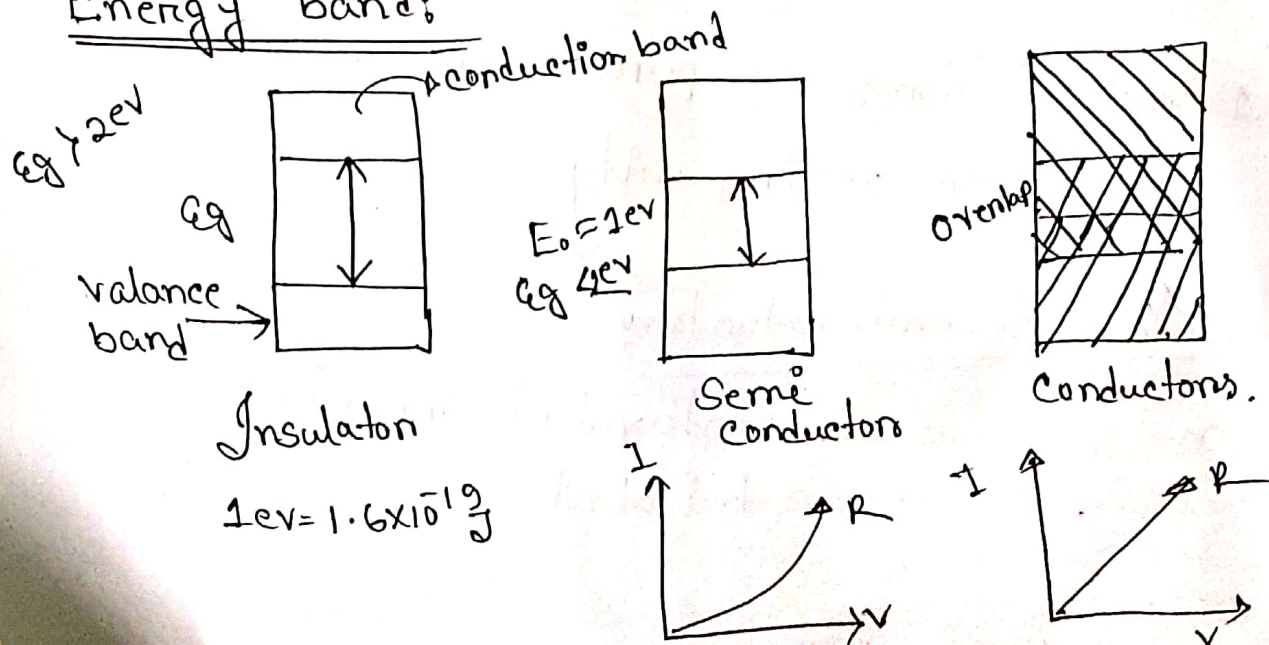
Semiconductor materials:

Conductor: The term conductor is used for any material that will support a generous flow of charge when the voltage is applied across its terminals
ex: Cu, Al

Insulator: An insulator is a material that offers very low conductivity when voltage is applied.
ex - mica, wood.

Semiconductor: It is a material that has conductivity more than insulators but less than conductor.
ex - Si, Ge

Energy band:



Intrinsic & Extrinsic Semiconductors:

Intrinsic Semiconductors:

They are pure semiconductors
free electrons are only due to natural causes

Extrinsic Semiconductors:

- Impurity atoms are added.
- Two types of impurities are there.
- Added 1 part in 10 million.
- Process of adding certain impurity atoms to pure semiconductor is called doping.

(i) Pentavalent: Sb, P, As (Group 5)

(ii) Trivalent: Boron, Gallium, Al, Indium (Group 3)

10 মিলিয়ন atoms pure / Intrinsic semiconductor
আমরা 1 impurity atom মাজ করি।

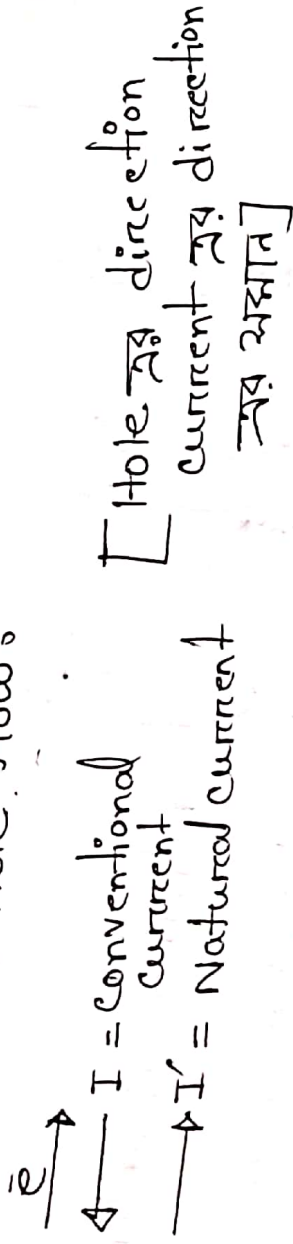
10 million atoms of pure semiconductor we
add 1 atom of impurity

Extrinsic Semiconductors:

n-type: we add pentavalent impurity

p-type: " " trivalent "

☐ Electron vs Hole flow:



number of electron = no. of hole (Intrinsic semiconductor)

* Under thermal equilibrium $\Rightarrow G = R$

☐ Generation: Temperature increases electron

Valence shell takes conduction band and takes

energy e^- and generates hole and energy is

☐ Recombination: When semiconductor is fixed

temp increases valence shell takes energy

electron valence shell and gives energy hole

and generates energy

Properties of Semiconductor:

1. The resistivity of semiconductor lies between a conductor and an insulator.

2. At 0K it behaves as insulator

3. If we increase the temperature of semiconductor, its electrical conductivity also increases.

4. They have an empty conduction band and almost filled valance band ok.
5. They are formed by a covalance bond
6. variable resistance ~~27~~ (electronics)
7. They have small energy gap.

Mass action law:

$$n = p = n_i \rightarrow \text{Intrinsic carrier density}$$

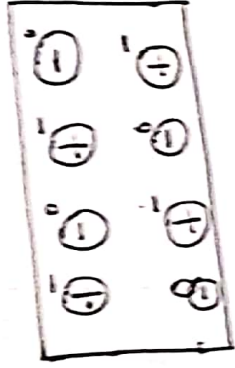
statement: under thermal equilibrium the product of free electron concentration and the free hole concentration is equal to a constant to the square of intrinsic carrier concentration.

$$\boxed{n \cdot p = n_i^2}$$

$R \propto n$	$R \propto p$	$G = n n_i^2$
$R \propto n p$		$G = R$
$R = n n p$		$n n_i^2 = n n p$
		$n_i^2 = n p$

[H] charge density in a semiconductor

Electrical neutrality



$N_D =$ concentration of donor atoms

$N_A = "$ " acceptor "

$p =$ hole con.

$n = e^-$ concentration

$$N_D + p = N_A + n \quad \text{--- (1)}$$

* In n-type semic-

$$N_A = 0$$

from (1)

$$N_D + p = n$$

$$N_D = n - p$$

$$N_D \approx n$$

$$n \approx N_D$$

$\rightarrow e^-$ concentration in a n-type material

$$np = n_i^2$$

$$N_D p = n_i^2$$

$$\rightarrow \boxed{p_n = \frac{n_i^2}{N_D}}$$

$$\text{In p-type}$$

$$N_D = 0$$

$$N_A + n = p$$

$$N_A = p - n \quad | \quad p \gg n$$

$$N_A \approx p$$

$$p \approx N_A$$

$$p_p \approx N_A$$

$$n_p p_p = n_i^2$$

$$n_p N_A = n_i^2$$

$$\boxed{np = \frac{n_i^2}{N_A}}$$

[H] conductivity of semiconductor

$$V_d = \mu E \quad \rightarrow \text{electric field intensity}$$

$$\quad \quad \quad \rightarrow \text{mobility}$$

$$\quad \quad \quad \rightarrow \text{velocity}$$

$$\text{charge density } J = qE$$

$$\boxed{J = ne\mu E}$$

$$J = (n\mu_n + p\mu_p) eE$$

$$G = (n\mu_n + p\mu_p)e$$

Ex 1 The mobilities of free $\bar{e}$ and h in a pure germanium are 0.38 and $0.18 \text{ m}^2/\text{V}\cdot\text{s}$. Find the value of intrinsic conductivity. Assume $n_i = 2.5 \times 10^{14}/\text{m}^3$ at room temperature.

$$\mu_n = 0.38 \text{ m}^2/\text{V}\cdot\text{s}$$

$$\mu_p = 0.18$$

$$\text{exp } n = p = n_i = 2.5 \times 10^{14} \text{ m}^{-3}$$

$$G = (0.38 + 0.18) \times 2.5 \times 10^{14} \times 1.6 \times 10^{-19}$$

$$= 0.56 \times 4$$

$$= 2.24 \text{ A/m}$$

Ex 2 In a germanium sample, a donor-type impurity is added to extent of 1 atom per 10^8 germanium atoms. Show the resistivity of the germanium sample drops to 3.7 ohm-cm .

Given:

$$\mu_n = 3800 \text{ cm}^2/\text{V}\cdot\text{sec}$$

$$\mu_p = 1800 \text{ cm}^2/\text{V}\cdot\text{sec}$$

$$q = 1.602 \times 10^{-19} \text{ coulombs}$$

$$n_i = 2.5 \times 10^{13}/\text{cm}^3$$

$$N_d = 4.41 \times 10^{12}/\text{cm}^3$$

Solution:

$$\text{resistivity } (\rho) = \frac{1}{G}$$

$$G = (n\mu_n + p\mu_p)q$$

$$ND = \frac{nq}{10^8}$$

$$= \frac{4.41 \times 10^{12}}{10^8}$$

$$= 4.41 \times 10^{14} \text{ cm}^{-3}$$

$$n \approx ND$$

$$n = 4.41 \times 10^{14}/\text{cm}^3$$

$$np = n_i^2$$

$$NDp = n_i^2$$

$$p = \frac{n_i^2}{ND} = \frac{(2.5 \times 10^{13})^2}{4.41 \times 10^{14}} = 1.41 \times 10^{12} \text{ cm}^{-3}$$

$$\begin{aligned}
 G &= (n\mu_n + p\mu_p)q \\
 &= (4.41 \times 10^{14} \times 3800 + 1.4172 \times 10^{12} \times 1800) \times 1.602 \times 10^{-19} \text{ (A-cm)}^{-1} \\
 &= 26884.18 \times 10^{-5} \\
 &= 0.26884 \text{ (A-cm)}^{-1}
 \end{aligned}$$

$$S = \frac{1}{0.26884} \text{ A-cm}$$

$$= 3.719 \text{ A-cm}$$

Consider an intrinsic Silicon bar of cross-section 8 cm^2 and length 0.5 cm at room temperature & an average field of 20 V/cm is applied across the ends of the Silicon bar.

- Electron and hole components of current density?
- Total current in the bar?
- Resistivity of the bar?

$$\mu_n = 1400 \text{ cm}^2/\text{V-sec}$$

$$\mu_p = 460 \text{ cm}^2/\text{V-sec}$$

$$q = 1.602 \times 10^{-19} \text{ C}$$

$$n_i = 1.5 \times 10^{10} / \text{cm}^3$$

$$\textcircled{a} \quad n = p = n_i$$

$$E = 20 \text{ V/cm}$$

$$J = n\mu_n E + p\mu_p E$$

electron component of current density $\rightarrow n\mu_n$

$$n\mu_n E = 1.5 \times 10^{10} \times 1400 \times 1.602 \times 10^{-19} \times 20$$

$$= 67.20 \times 10^{-6} \text{ A/cm}^2$$

$$= 67.20 \mu\text{A/cm}^2$$

$$p_{Kp} q E = 1.5 \times 10^{10} \times 450 \times 1.601 \times 10^{-19} \times 20$$

$$= 21.6 \mu A/cm^2$$

(b)

$$J = i/A$$

$$i = JA$$

$$= (67.20 + 21.6) \times 5 \mu A$$

$$= 444 \mu A$$

(c)

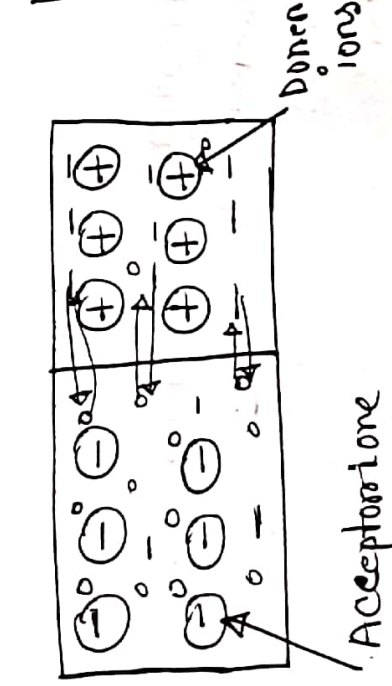
$$f = \frac{1}{6}$$

$$= \frac{1}{(n_{Kp} + p_{Kp}) q}$$

$$f = 22.52 \times 10^4 \text{ cm}$$

PN Junction Diode:

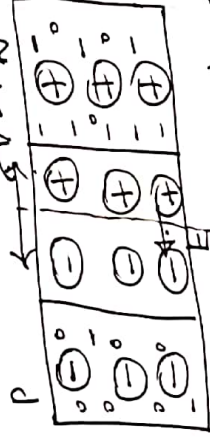
Bias: Application of external voltage across the two terminals.



Diffusion current: Majority charge carriers movement from high concentration to low concentration.

Depletion region: because of diffusion recombination took place and immobile ions will surface out.
 • NO mobile charge but only uncovered immobile ions

immoblie ions



diffusion current
drift current

Drift current = minority charge carriers

④ under steady state $\rightarrow$ drift current = diffusion current
 Diffusion current = drift current

⑤ The net current = 0 for open circuit diode

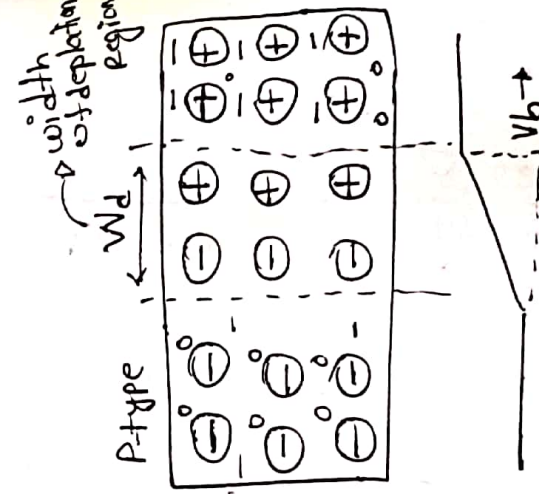
Barrier Potentials

$$V_b = \frac{kT}{e} \ln \left(\frac{N_A N_D}{n_i^2} \right)$$

$$V_T = \frac{kT}{e} \rightarrow 1.38 \times 10^{-23} \text{ J/K} \rightarrow 1.16 \times 10^{-19} \text{ eV}$$

$$V_T = 0.026 \text{ volts} \rightarrow \text{valid at room temp. only}$$

$$V_T = \frac{273 + T(^{\circ}\text{C})}{11600} \rightarrow \text{for any temp.}$$



Width of Depletion Region

$$W_d = \sqrt{\frac{2\epsilon}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) V_b} \quad \leftarrow \text{Carrier Potential}$$

ϵ = electrical permittivity

for Si

$$\epsilon = 1.04 \times 10^{-12} \text{ F/cm}$$

$$\epsilon = \epsilon_0 \epsilon_r$$

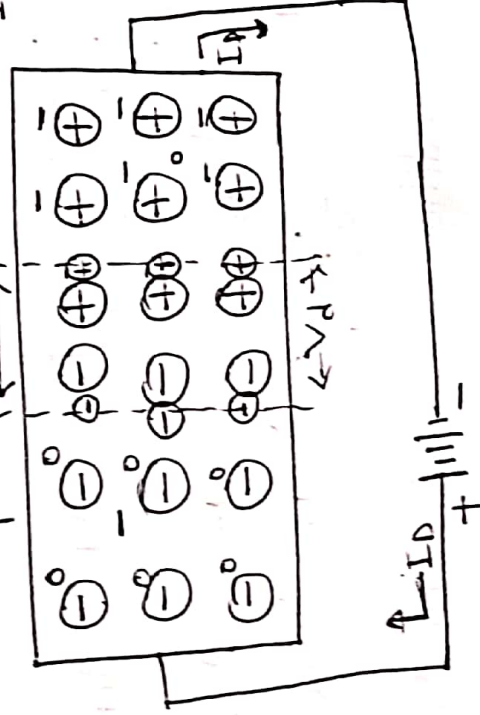
ϵ_r = Permittivity of
Relative Vacuum
Permittivity

$$\epsilon_r = 1.2448 \times 10^{-12} \text{ F/cm}$$

$$q = 1.6 \times 10^{-19} \text{ C}$$

Forward-Bias Condition $\xrightarrow{\text{Imaginary current}}$

P $\leftarrow W_d$, N $\leftarrow$ Minority current



$$I_D = I_{maj} - I_S$$

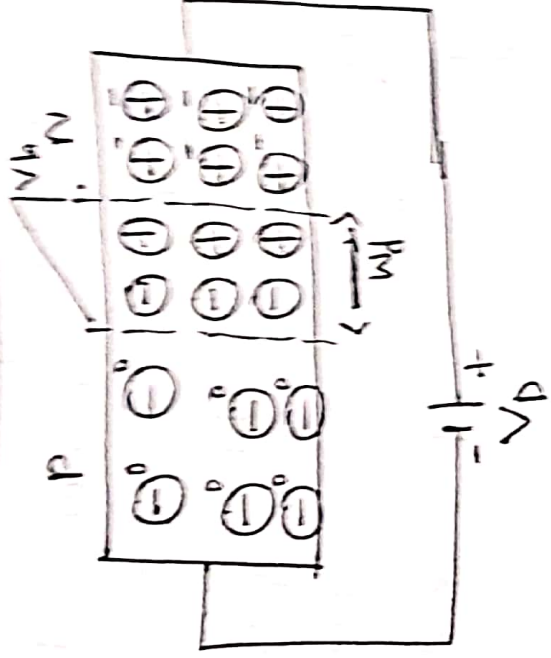
diffusion current drift current

যখন V_D বা ϕ_{bi} potential বৃদ্ধি করলে তখনই P-সাইডে $\ominus$ চার্জের $\oplus$ প্রাক্তন ও n-সাইডে চার্জের $\ominus$ প্রাক্তন থাকবে। P-সাইডের hole প্রাক্তন $\ominus$ immobile অ্যানুনের সাথে যুক্ত হয়ে N side-এর electron + immobile অ্যানুনের সাথে Recombination ঘটবে।

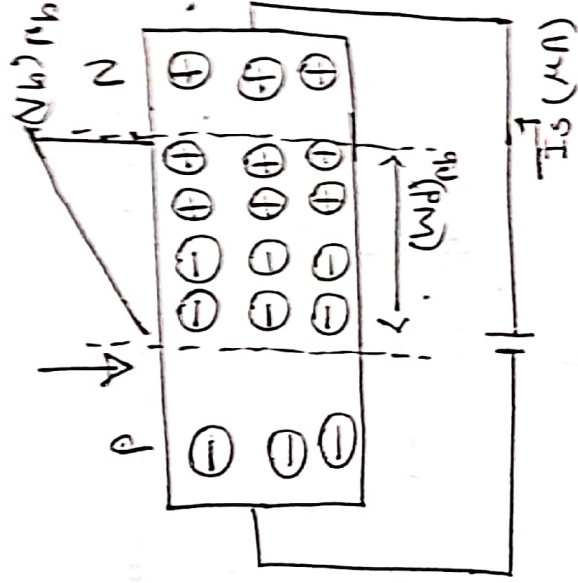
ফলত depletion region কমে যায়।

$$V_D \uparrow W_d \downarrow \quad \leftarrow \text{New barrier potential} = V_D - V_A$$

Reverse Bias Condition



For no bias condition
 Barrier potential $= V_b$
 width $d \propto \sqrt{V_b}$



$$(V_b)_{nb} = V_b + V_D$$

$$V_D \uparrow (V_b)_{nb} \uparrow$$

$$V_D \uparrow (W_D)_{nb} \uparrow$$

$$(W_D)_{nb} > (W_D)_{nb} > (W_D)_{nb}$$

$$(V_b)_{nb} > (V_b)_{nb} > (V_b)_{nb}$$

বিভিন্ন বাহ্যিক condition & depletion region এর

সহ majority charge carriers move, বহুদ্রব্য

$$\text{So } I_{maj} \approx 0$$

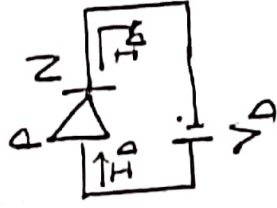
$$\text{এবং } I_{min}$$

Semiconductor Diode:

trivalent → pentavalent

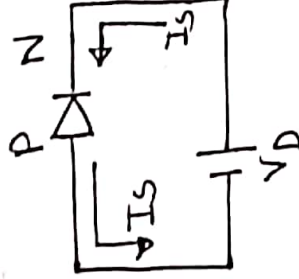


forward bias:



$$I_D = I_{mg} - I_S$$

Reverse bias:



Diode current and voltage relation:

$$I_D = I_S \left(e^{\frac{kV_D}{T_k}} - 1 \right)$$

I_D = Diode Current

I_S = Reverse Saturation current

V_D = Voltage across the diode.

T_k = Temp in K

$$k = 1.38 \frac{11600}{n}$$

n = Ideality factor

$\left[\begin{matrix} n=1 & \text{for Ge} \\ n=2 & \text{for Si} \end{matrix} \right]$ I_D low

$n=1$ for Ge & Si I_D high

$$V_T = \frac{T_k}{11600}$$

$$V_T = \frac{T_k}{k n}$$

$$\frac{k}{T_k} = \frac{1}{V_T n}$$

$$I_D = I_S \left(e^{\frac{V_D}{n V_T}} - 1 \right)$$

Q A Germanium diode displays a forward voltage of 0.25V at 10mA current at room temperature. Find the reverse saturation current.

$$V_D = 0.25V \quad V_T = \frac{T_K}{11600} = \frac{300}{11600} = 0.026V$$

$$I_D = 10mA = 10 \times 10^{-3}A$$

$$T = 300K$$

$$I_S = ?$$

$$I_D = I_S \left(e^{\frac{V_D}{nV_T}} - 1 \right) \quad n=1$$

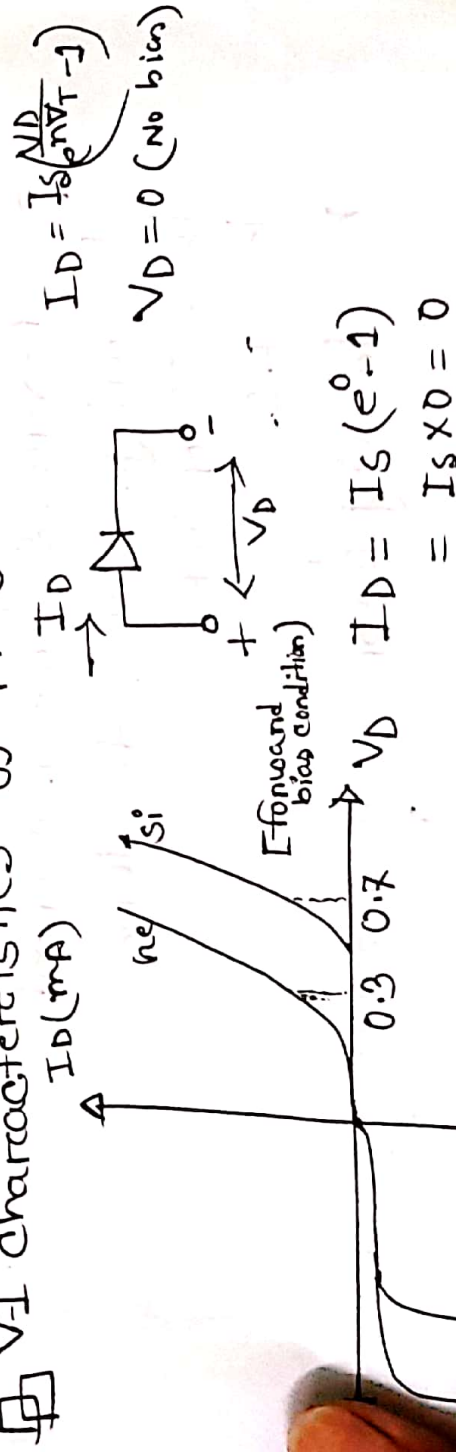
$$10 = I_S \left(e^{\frac{0.25}{1 \times 0.026}} - 1 \right)$$

$$10 = I_S \cdot 14986.922$$

$$I_S = \frac{10}{14986.922} = 6.67 \times 10^{-7} mA$$

$$= 6.67 \times 10^{-7} A$$

Q VI characteristics of PN junction Diode:-



$$V_D \uparrow \quad W_D \downarrow \quad V_B \downarrow$$

V_D Barrier potential I_n forward bias we will get current $V_D = V_B$ we will get current

get current flow (high level) $I_D = I_{max} - I_S$

Reverse bias condition



$$I_D = I_S (e^{\frac{V_D}{kT}} - 1)$$

$$I_D = I_S (-1)$$

$$I_D = -I_S (\text{Reverse current})$$

$$I_D = I_S (I_{maj} = 0) \quad \text{carrier is fixed.}$$

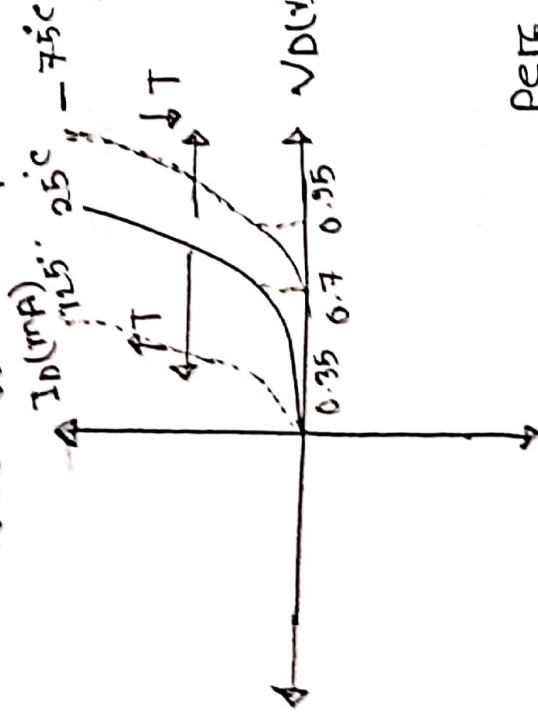
Number of minority charge.

At voltage V_D under reverse bias condition

high level p -region current and other Break down voltage/Zener voltage.

Effect of temperature on V-I characteristics:

forward bias condition:-



In case of forward bias region of si diode shift to the left at a rate of 2.5 mV per degree centigrade rise in temperature.

EX: $25^{\circ}\text{C} \rightarrow V_D = 0.7 \text{ V}$
for 100°C $\uparrow$ in temp on at 125°C $V_D = 2$

$$100 \times 0.25 \text{ V}$$

$$V_D \downarrow \text{ by } 0.25 \text{ V}$$

$$\text{New } V_D = (0.7 - 0.25) \text{ V} \\ = 0.45 \text{ V}$$

for reverse bias condition:-

I_s double for every